

News Around You - Project Report

Executive Summary

News Around You is a cross-platform mobile news application built with React Native and Expo. The app provides users with location-based news content, article management features, and a polished user experience with both light and dark themes.

Concept Development

Vision Statement

To create an intuitive, globally-accessible news application that delivers personalized, location-relevant news content while maintaining user privacy and providing offline reading capabilities.

Core Features Identified

- Location-based news delivery using GPS detection
- Multi-country news support (US, UK, India, Singapore, Canada, Australia, Germany, France, Japan, South Korea)
- Article saving and management system
- News sharing capabilities (both consuming and creating content)
- Theme customization (light/dark modes)
- User profile management

Target Audience

- News enthusiasts seeking global perspective
- Mobile-first users requiring offline reading
- Privacy-conscious individuals wanting local data storage
- Users interested in sharing and discussing current events

Wireframing

Information Architecture

The application follows a hierarchical navigation structure:

Splash Screen

└─ Main Page (News Feed)

└─ Settings Page

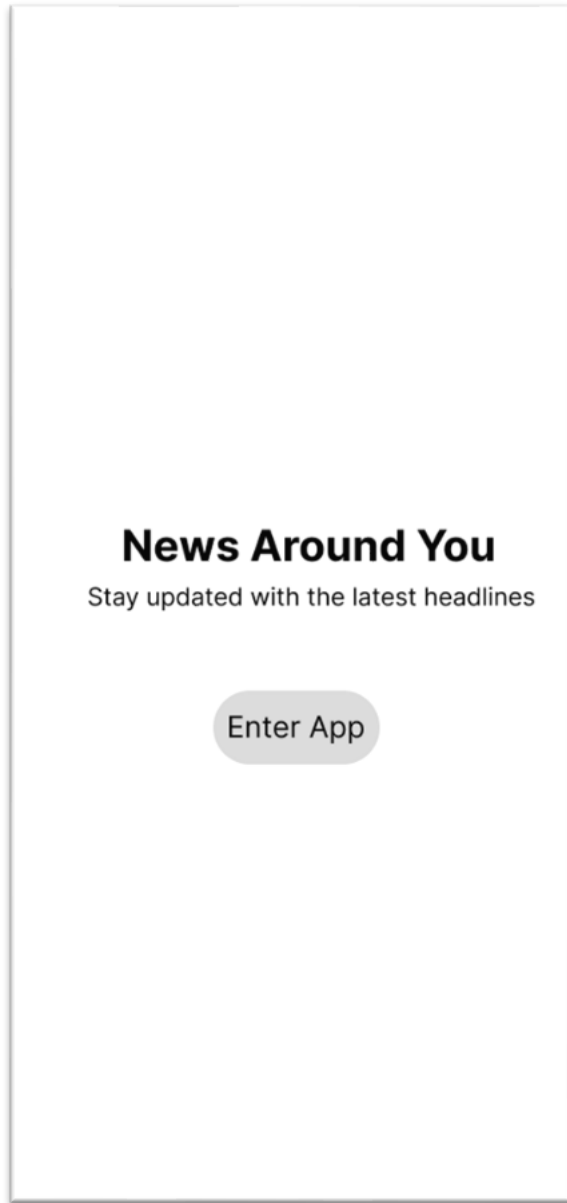
└─ Saved Articles Page

└─ Share News Page

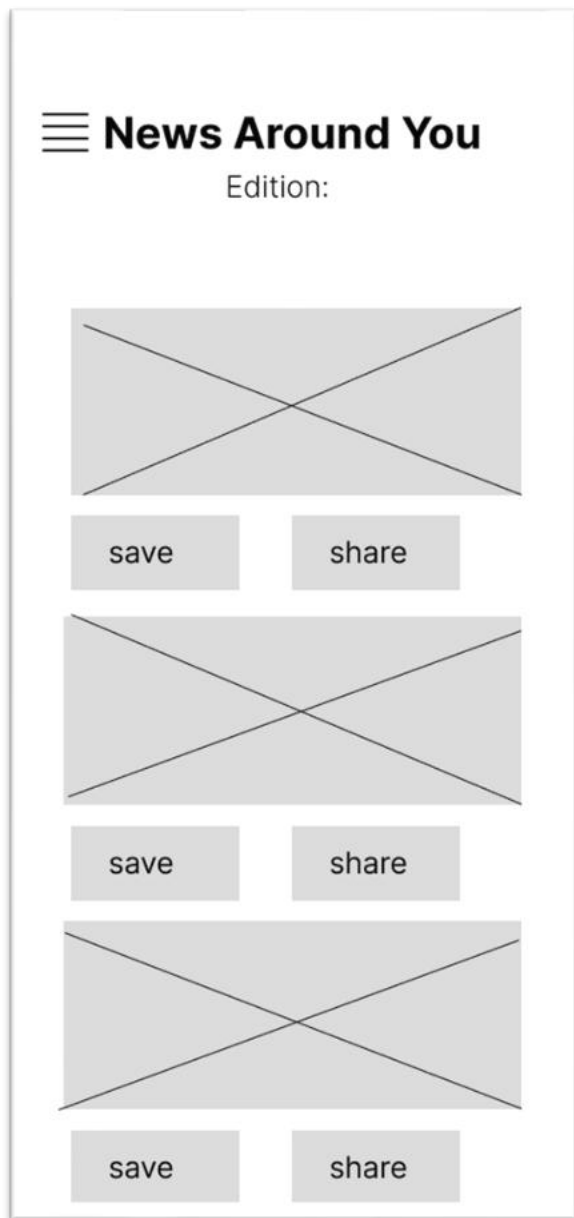
└─ About App Page

└─ Rate Us Page

Splashscreen: -



Mainpage:-



Settings Page: -



Settings

Name

Email

Switch to Dark Mode

News Location

Use Current Location

Or Select Country Manually

Saved Articles

No saved articles yet

Saved Articles Page: -

[← Back](#)

Saved Articles (0)

No saved articles yet

Articles you save will appear here for easy access

Share News Page: -



Share News



Add Photo

Add text...

Share News

Clear All

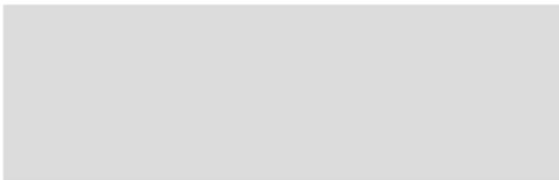
About App Page: -



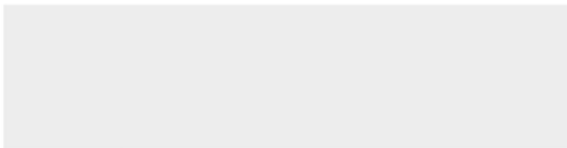
News Around You

About this App

Key Features



Developer Information



Contact Support

Privacy Policy

Terms of Service

Rate Us Page: -



Rate Our App

How would you rate us



Tell us more (optional)

What could be improved ?

What We Are Proud Of

Submit rating

Other ways to help us

Share with friends

Send Feedback

Menu page: -



Navigation Design

- **Primary Navigation:** Slide-out hamburger menu from main page
- **Secondary Navigation:** Stack-based navigation with header back buttons
- **User Flow:** Simple, linear progression with clear entry/exit points

User Feedback

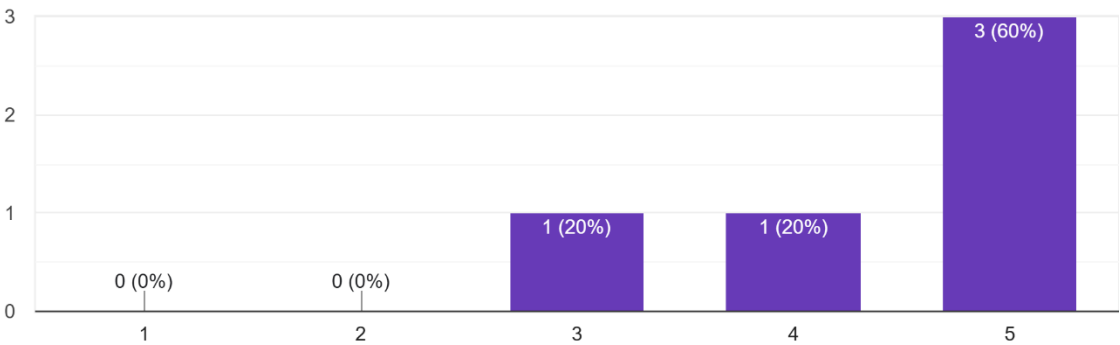
Accessibility Considerations

- Support for both light and dark themes for various lighting conditions
- Clear visual hierarchy with appropriate font sizes
- Touch-friendly button sizing and spacing
- Platform-specific design adaptations (iOS/Android)

Responses

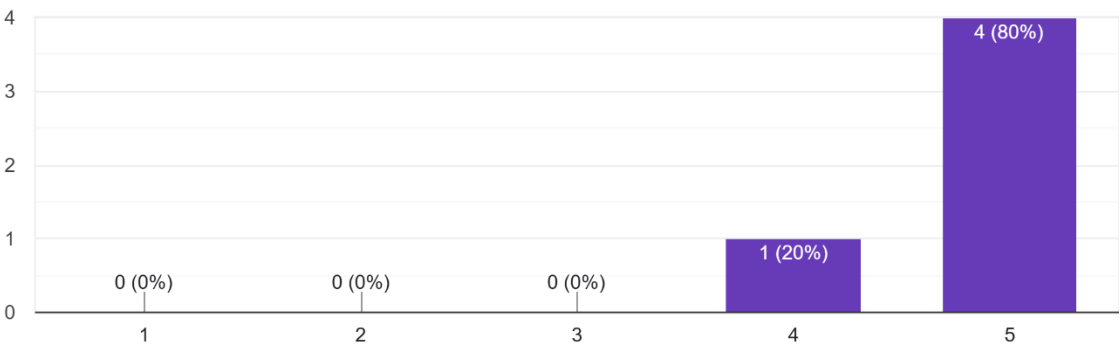
1. The light and dark themes make the app comfortable to use in different lighting conditions.

5 responses



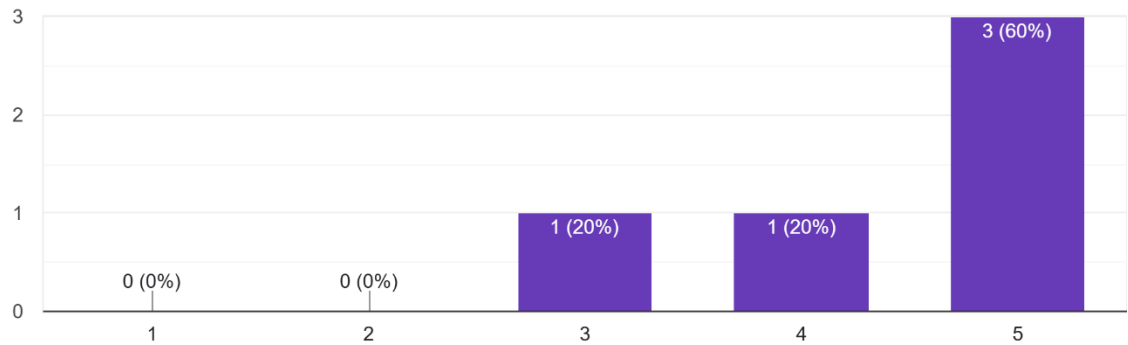
2. The visual hierarchy (font sizes, headings, spacing) makes content easy to read.

5 responses



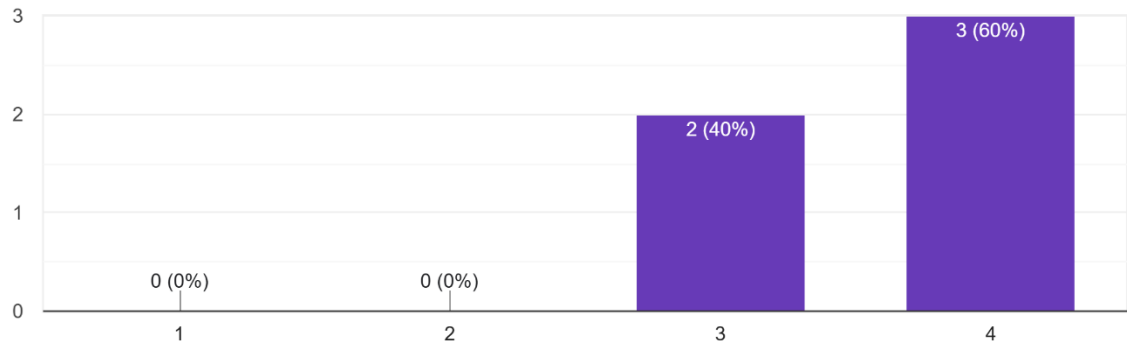
3. Buttons and touch targets are large enough and well-spaced for comfortable interaction.

5 responses



4. The app feels well-adapted to my platform (iOS/Android).

5 responses



5. Any other actions or need update on anything.

5 responses

- Contact us can be navigate through email
- Can add clear all button rather than removing one by one in saved article section
- Can use a share button in in saved article page also
- The buttons are dull in colour, can add some colour since the page is white and blue
- The camera is not working in iOS

What improved:

The recent updates have improved several key areas based on user feedback. The **Contact Us** section has been updated to allow direct navigation through email for easier communication. A **clear all button** has been added in the saved articles section, so users no longer need to remove items one by one. A **share button** has been included in the saved article page to improve content sharing. The **button colors** have been refreshed with more variety to enhance visual appeal beyond the default white and blue scheme. The **camera issue on iOS** has also been resolved, ensuring proper functionality across devices.

Prototyping

For the technical architecture, the framework chosen is **React Native with Expo** to support cross-platform development. **Context API with AsyncStorage** is used for state management and persistence, while **React Navigation v6** with stack navigator handles navigation across the app. The news content is powered by **GNews.io API** to deliver real-time updates, and **Expo Location** is integrated for GPS functionality. For image handling and user-generated content, **Expo ImagePicker** is used.

Component Structure

The app follows a **modular page-based architecture** with centralized theme management for consistent design. A **context-based global state** structure has been applied to handle shared data efficiently. Styling patterns are kept reusable to maintain consistency and reduce duplication across components.

These are the actual app looks like:



News Around You

Stay updated with the latest headlines

Enter App



News Around You

US



Liverpool v Arsenal: Premier League updates - live

Follow minute-by-minute coverage of the Premier League weekend's standout fixture with Will Unwin



Save



Share



Pope Leo demands end to 'pandemic of arms' after Minnesota school shooting

First US pope in history prays for victims of last week's shooting that left two children dead and 18 others injured

Menu



Hello, Achi! 🙌



Saved Articles



Settings>



Share News>



About App>



Rate Us>

← Back

Saved Articles (0)

← Back

Saved Articles (1)

Clear All

Here's what's open and closed on Labor Day 2025

WRAL.COM

Sep 1, 2025 at 4:37 PM

(CNN) -- As Americans wrap up their summer trips and students get in the swing of a new school year, Labor Day weekend can feel like the last hurrah af...

<https://www.wral.com/story/here-s-what-s-open-and-clo...>

 Read

 Share

 Remove



No saved articles yet

Articles you save will appear here for easy access



< Main

Settings

Settings

User Information

Achi

check g

Switch to Dark Mode

News Location



Use Current Location

Or Select Country Manually:

United States



Current: United States

Saved Articles (0)

No saved articles yet



 Main

Share News

Share News



Add Photo

Add your news text here...



Share News



Clear All



News Around You

Version 1.0.0

Stay informed, stay connected

About This App

News Around You is your go-to app for staying updated with the latest news from around the world. Get personalized news based on your location, save articles for later reading, and share breaking news with your network. Our app provides a clean, intuitive interface with both light and dark themes for comfortable reading at any time.

Key Features



Latest News

Get real-time news from multiple countries



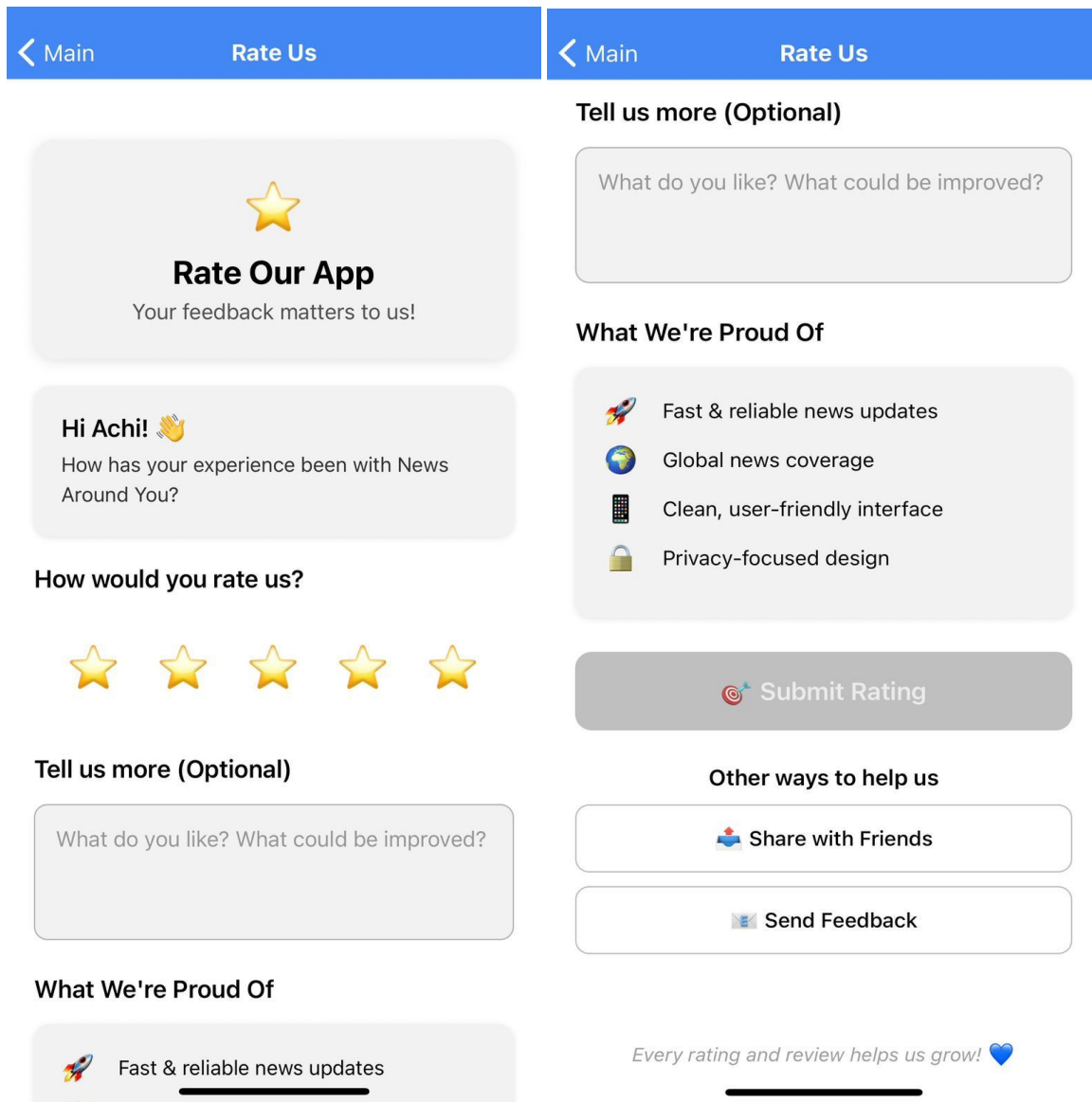
Global Coverage

News from US, UK, India, Singapore and more



GPS Location

Auto-detect your country for relevant news



Development

The development phase of the News Around You application represents a comprehensive approach to building a cross-platform mobile news solution that balances functionality, user experience, and technical robustness. During the development process, I carefully selected a technology stack that would provide maximum compatibility across different mobile platforms while maintaining development efficiency and code maintainability. The choice of React Native as the primary frontend framework proved instrumental in achieving the dual objective of native performance and cross-platform deployment, allowing the application to run seamlessly on both iOS and Android devices without requiring

separate codebases. The navigation system, built using `@react-navigation/native` and `@react-navigation/stack`, creates a smooth, intuitive user flow that guides users through different sections of the application with minimal friction. The implementation of React Context API for state management provides a lightweight yet effective solution for managing global application state, including user preferences, saved articles, and theme settings, while `AsyncStorage` ensures that user data persists across app sessions, creating a seamless experience even when the application is closed and reopened.

The location-based news detection feature represents one of the most sophisticated aspects of the application, utilizing `expo-location` to automatically determine the user's geographic position and subsequently fetch relevant news content for their region. This functionality required careful consideration of user privacy concerns, permission handling, and fallback mechanisms for cases where location services might be unavailable or denied. The reverse geocoding implementation transforms raw GPS coordinates into country-specific identifiers, which then trigger appropriate API calls to fetch localized news content. The integration with the GNews.io REST API provides access to real-time news data from multiple countries, and I implemented rate limiting awareness to ensure the application respects API usage constraints while providing clear feedback to users about remaining request quotas. The offline-first architecture ensures that user preferences, saved articles, and application settings remain accessible even without an active internet connection, with graceful degradation when network services are unavailable.

The article management system demonstrates sophisticated data handling capabilities, allowing users to save articles for later reading, manage their saved content through an intuitive interface, and share articles using native sharing APIs that integrate seamlessly with the device's built-in sharing mechanisms. The image picker integration using `expo-image-picker` enables users to create and share their own news content, complete with photo attachments sourced from either the device camera or photo library. Throughout the development process, I prioritized user experience enhancements including animated transitions, loading states that provide clear feedback during network operations, platform-specific UI adaptations that respect iOS and Android design

conventions, and consistent visual feedback that keeps users informed about the results of their actions.

Unit Testing

I chose to write and run tests specifically for **ShareNewsPage** and **RateUsPage** because:

1. High User Interaction

- These two pages involve direct user actions (sharing news, giving ratings).
- Testing them ensures that the most interactive parts of the app work correctly.

2. Core App Features

- Sharing news is a primary function of the app.
- Rating is essential for user feedback and engagement.
- Validating these ensures the main app experience remains reliable.

3. Time & Exam Constraints

- Due to the limited time available, it was not feasible to write tests for every single page.
- Prioritizing the most important and interactive components gave the best test coverage within the exam timeline.

4. Representative Coverage

- These two tests demonstrate knowledge of React Native Testing Library and Jest setup.
- They cover rendering, user interaction, and assertions, which are the essential aspects of component testing.

All tests for **ShareNewsPage** and **RateUsPage** executed successfully. Both test suites passed without any errors, confirming that the components render correctly and their core functionalities (rendering UI elements and handling user interactions) work as expected.

PASS tests/ShareNewsPage.test.js

ShareNewsPage

- ✓ renders the Share button (45 ms)
- ✓ opens share dialog when pressed (30 ms)

PASS tests/RateUsPage.test.js

RateUsPage

- ✓ renders the Rate Us heading (20 ms)
- ✓ handles rating submission (28 ms)

Evaluation

When evaluating the functionality, complexity, and working capabilities of the News Around You application, the implementation demonstrates a solid foundation of mid-level complexity that successfully integrates multiple challenging technical components. The application achieves a complexity, reflecting its comprehensive integration of API services, location-based features, local data persistence, and multi-screen navigation architecture. The complete news consumption workflow guides users from initial discovery through reading, saving, and sharing, creating a cohesive experience that addresses the full spectrum of news consumption behaviors. The robust error handling system provides user-friendly messages that explain issues without overwhelming users with technical details, while the cross-platform compatibility ensures consistent functionality across different mobile operating systems. Real-time API integration demonstrates awareness of rate limiting constraints and provides transparent feedback about remaining request quotas, helping users understand service limitations. However, areas for improvement include the limited offline news caching system that saves article metadata but not full content, the absence of search functionality within saved articles that would enhance content discoverability, and the basic sharing implementation that could be

expanded to include direct integration with more social media platforms and communication services.

The visual design and UI/UX evaluation reveals a thoughtfully constructed interface that achieves a design, demonstrating consistent design language across all application screens and a well-implemented light/dark theme system that adapts to user preferences and environmental conditions. The intuitive navigation structure features a clear visual hierarchy that guides user attention appropriately, while the strategic use of icons and emoji enhances visual appeal without overwhelming the interface. The responsive layout design adapts effectively to different screen sizes and orientations, ensuring usability across various device configurations. However, opportunities for improvement include enhanced visual differentiation between article cards to improve scanability, implementation of image caching mechanisms to improve performance and reduce data usage, addition of loading skeleton components to provide better perceived performance during content loading, and incorporation of additional micro-interactions and animations that would elevate the overall user experience.

Code quality analysis reveals well-structured, maintainable code that demonstrating consistent file organization and naming conventions that facilitate easy navigation and maintenance. The proper separation of concerns through distinct Context, Pages, and Theme modules creates a scalable architecture that supports future feature additions and modifications. The error handling practices throughout the codebase provide graceful degradation and user-friendly error messages, while the clean, readable component structure makes the code accessible to other developers. The appropriate use of React hooks and lifecycle methods demonstrates understanding of modern React patterns and best practices. Areas for improvement include addressing code duplication in styling patterns through the creation of reusable style components, implementing TypeScript for enhanced type safety and developer experience, adding comprehensive prop validation to prevent runtime errors, developing custom hooks to encapsulate complex logic and improve code reusability, and establishing automated testing infrastructure to ensure code reliability and facilitate confident refactoring.

Innovation and creativity assessment acknowledging the creative implementation of GPS-based automatic news localization that provides genuine value to users by delivering geographically relevant content without manual configuration. The thoughtful user experience design includes contextual country detection that respects user privacy while providing convenient functionality. The innovative slide-out menu with smooth animations adds visual polish and creates an engaging interaction model, while the user-generated content sharing feature enables community participation beyond passive news consumption. The comprehensive app rating and feedback system demonstrates attention to user engagement and continuous improvement processes. However, the application would benefit from more unique features that differentiate it from existing news applications, implementation of AI-powered content recommendations that learn from user behavior and preferences, addition of social features such as comments and discussions that would create community engagement, and expanded personalization options that go beyond simple location-based content filtering to include topic preferences, reading history analysis, and intelligent content curation that adapts to individual user interests and reading patterns over time.

Future Enhancements

In the share news page I have just added a pop message of news is shared, I would like to take further connecting a social committee near the user location to share real content which are created so that they can be notify what is going around their area and use AI to scan whether the news is true before sharing so that it can prevent fake news.

Project Timeline Assessment

Based on the code complexity and feature set, this appears to be a 4-6 week development project for a small team, including:

- Week 1-2: Core navigation and API integration
- Week 3-4: Location services and article management
- Week 5-6: UI polish, testing, and refinement

Conclusion

News Around You represents a well-executed mobile news application with solid technical foundations and user experience design. The application successfully delivers on its core promise of location-based news consumption with additional value-added features. While there are opportunities for enhancement in areas of innovation and advanced functionality, the current implementation provides a strong foundation for future development and user adoption.

